Introduction to Computational Optics

Charles Yen

Why do we need computers?

- Simple vs Complex

$$F = ma$$

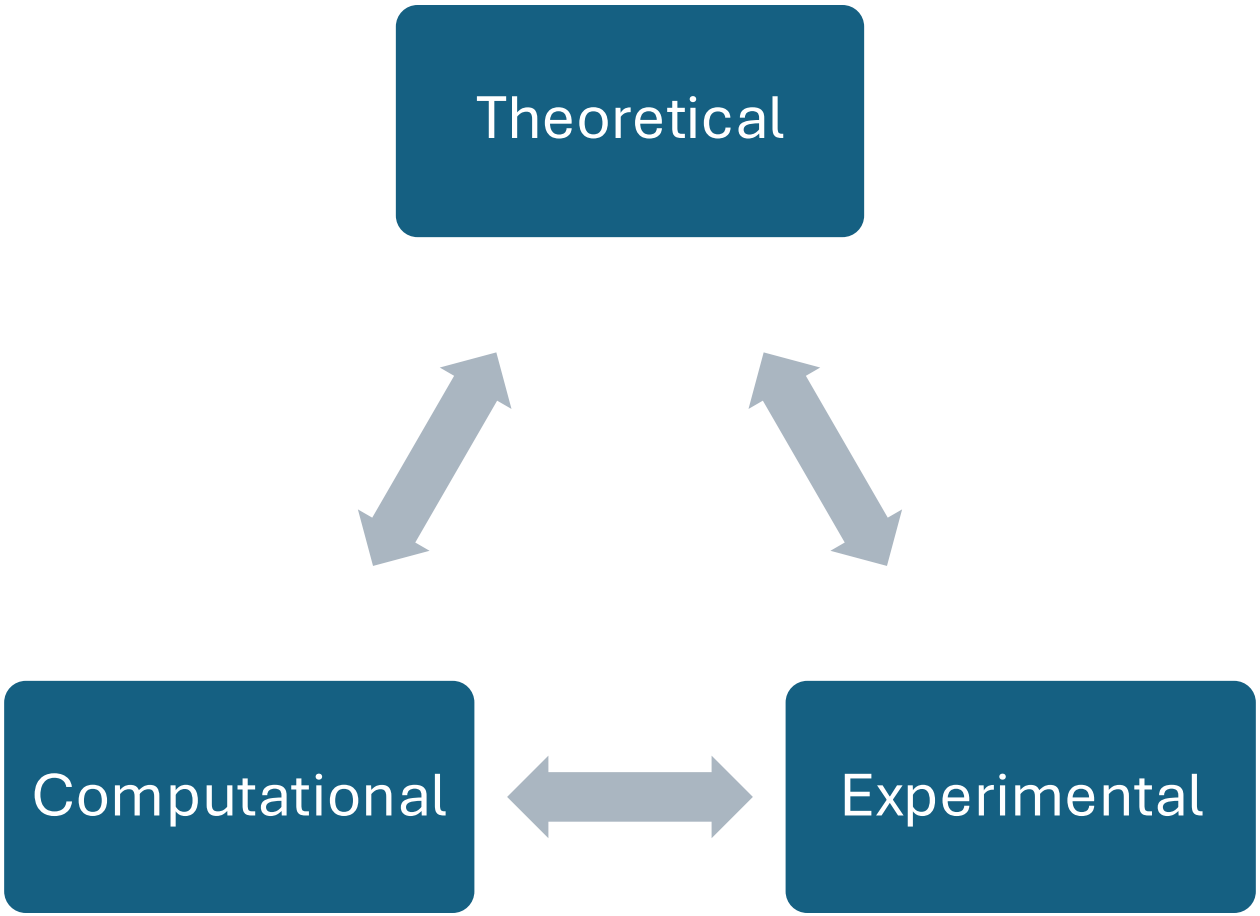
$$r : \rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\phi}{r}\frac{\partial u_r}{\partial \phi} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\phi^2}{r}\right) =$$
$$-\frac{\partial p}{\partial r} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_r}{\partial \phi^2} + \frac{\partial^2 u_r}{\partial z^2} - \frac{u_r}{r^2} - \frac{2}{r^2}\frac{\partial u_\phi}{\partial \phi}\right] + \rho g_r$$

$$\phi : \rho\left(\frac{\partial u_\phi}{\partial t} + u_r\frac{\partial u_\phi}{\partial r} + \frac{u_\phi}{r}\frac{\partial u_\phi}{\partial \phi} + u_z\frac{\partial u_\phi}{\partial z} + \frac{u_r u_\phi}{r}\right) =$$
$$-\frac{1}{r}\frac{\partial p}{\partial \phi} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_\phi}{\partial \phi^2} + \frac{\partial^2 u_\phi}{\partial z^2} + \frac{2}{r^2}\frac{\partial u_r}{\partial \phi} - \frac{u_\phi}{r^2}\right] + \rho g_\phi$$

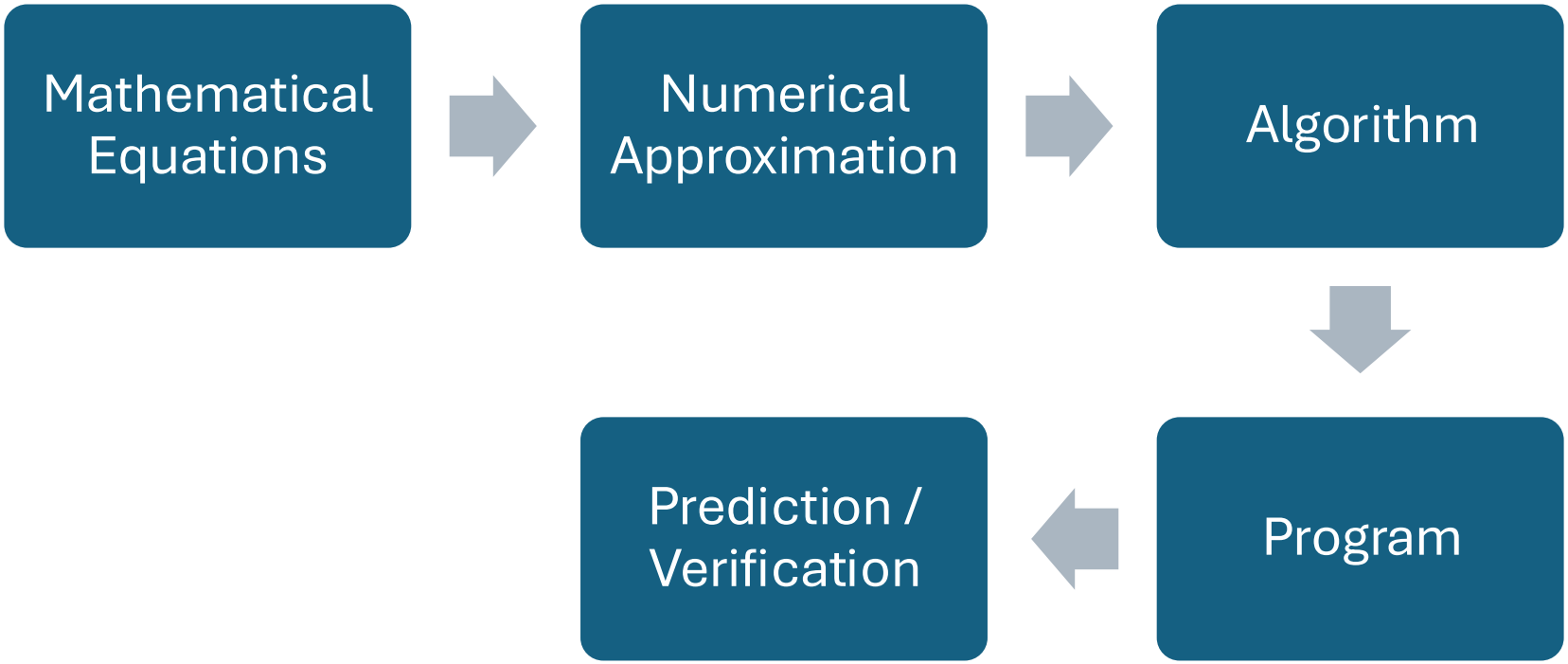
$$z : \rho\left(\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\phi}{r}\frac{\partial u_z}{\partial \phi} + u_z\frac{\partial u_z}{\partial z}\right) =$$
$$-\frac{\partial p}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \phi^2} + \frac{\partial^2 u_z}{\partial z^2}\right] + \rho g_z.$$



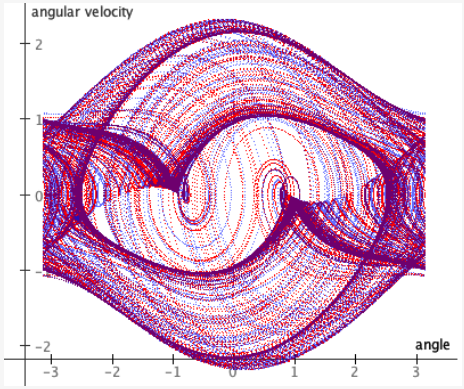
Pillar of Physics

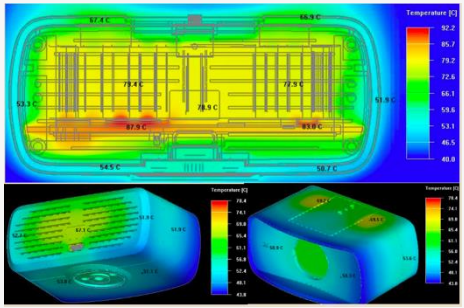


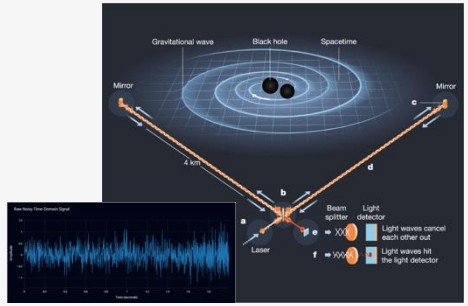
An Introduction to Computational Physics

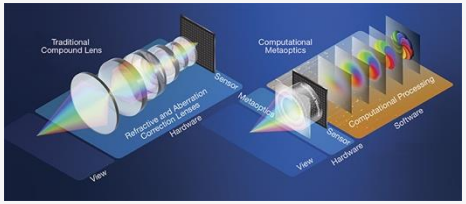


Samples that need simulations







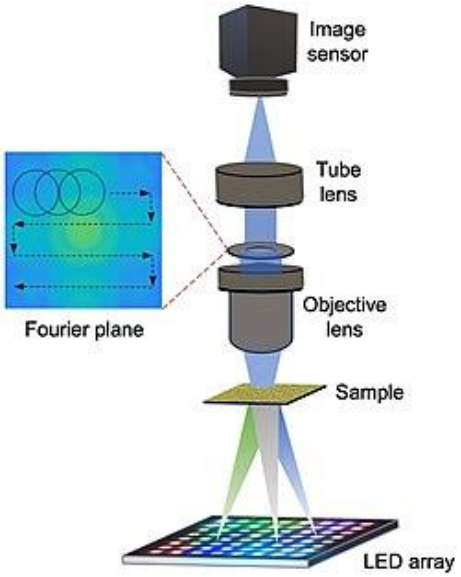


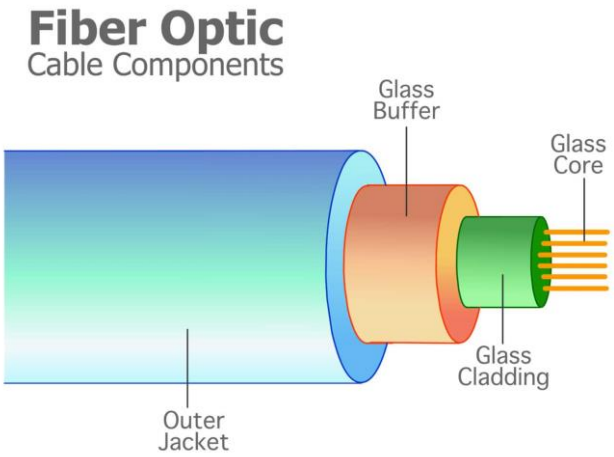
Why Computational Optics

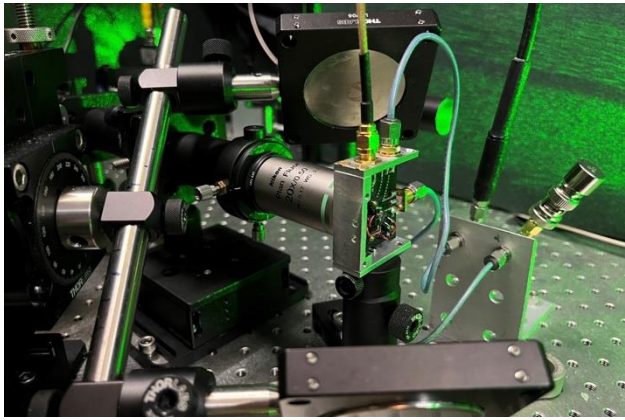
How do we design optical systems before building them?

Where to apply those subfields?

- Imaging
- Photonics
- Quantum Technologies

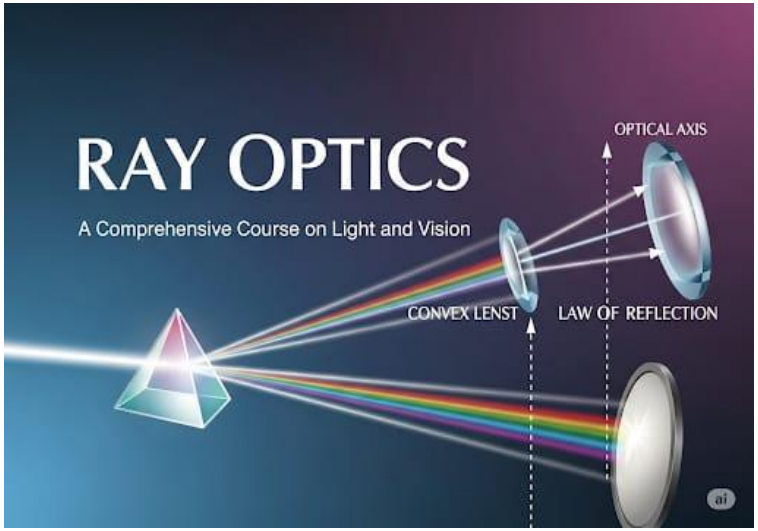


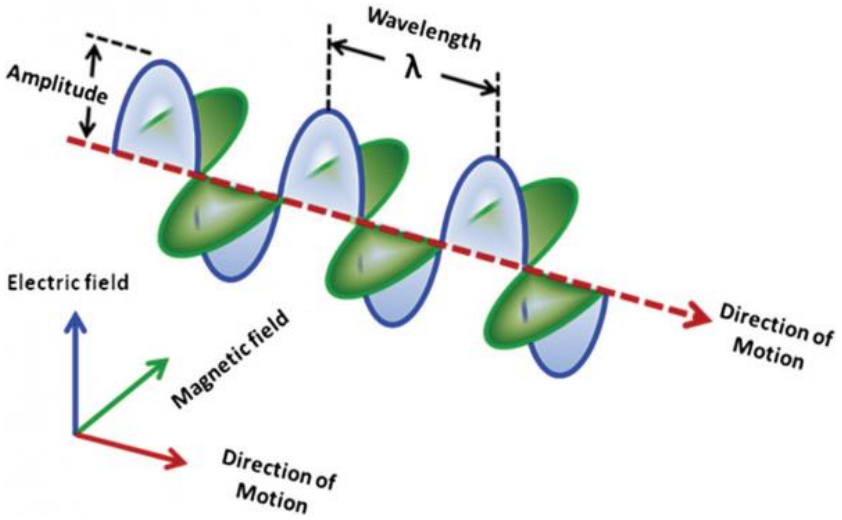




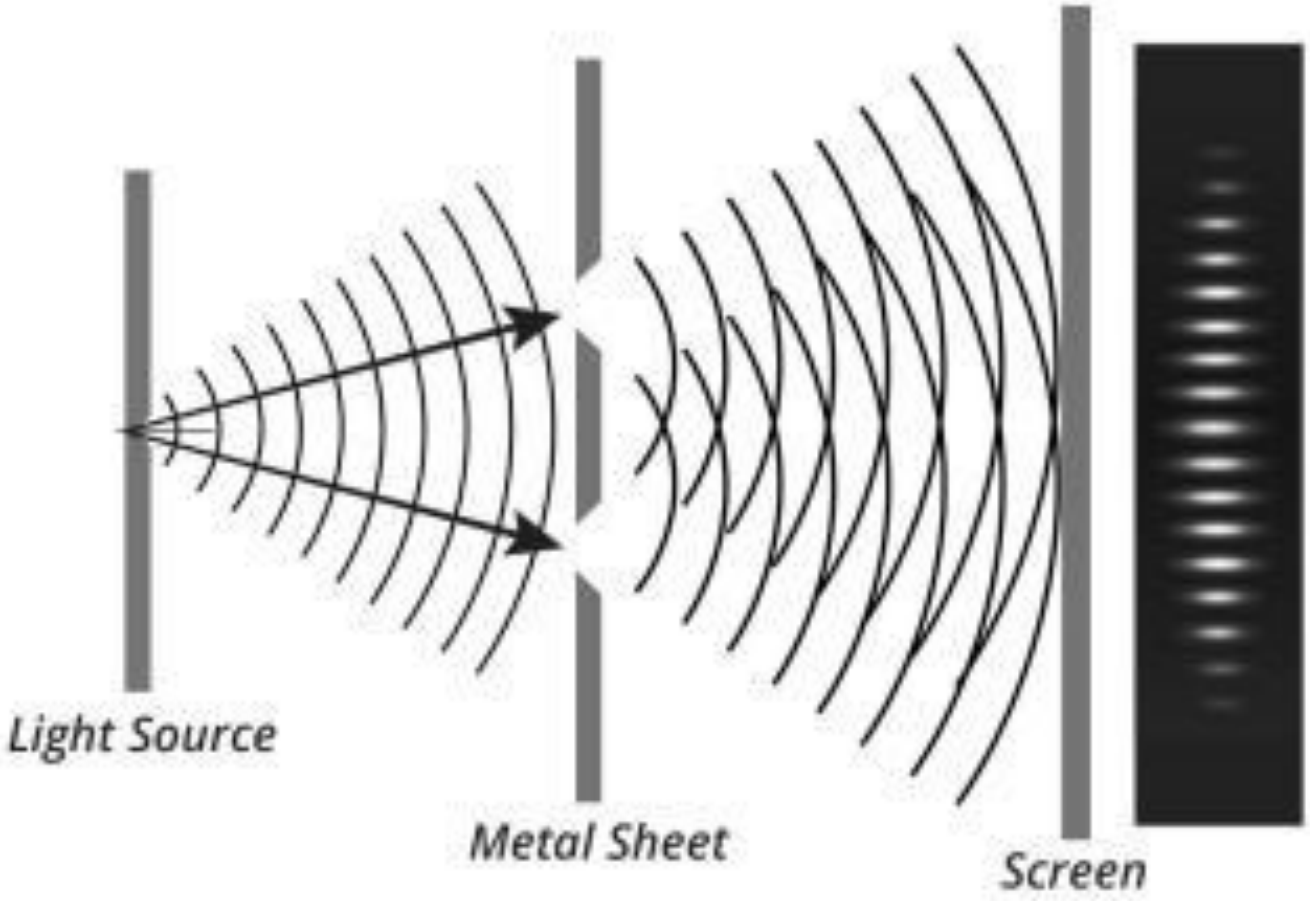
Two Models of Light

- Ray Optics
- Wave Optics

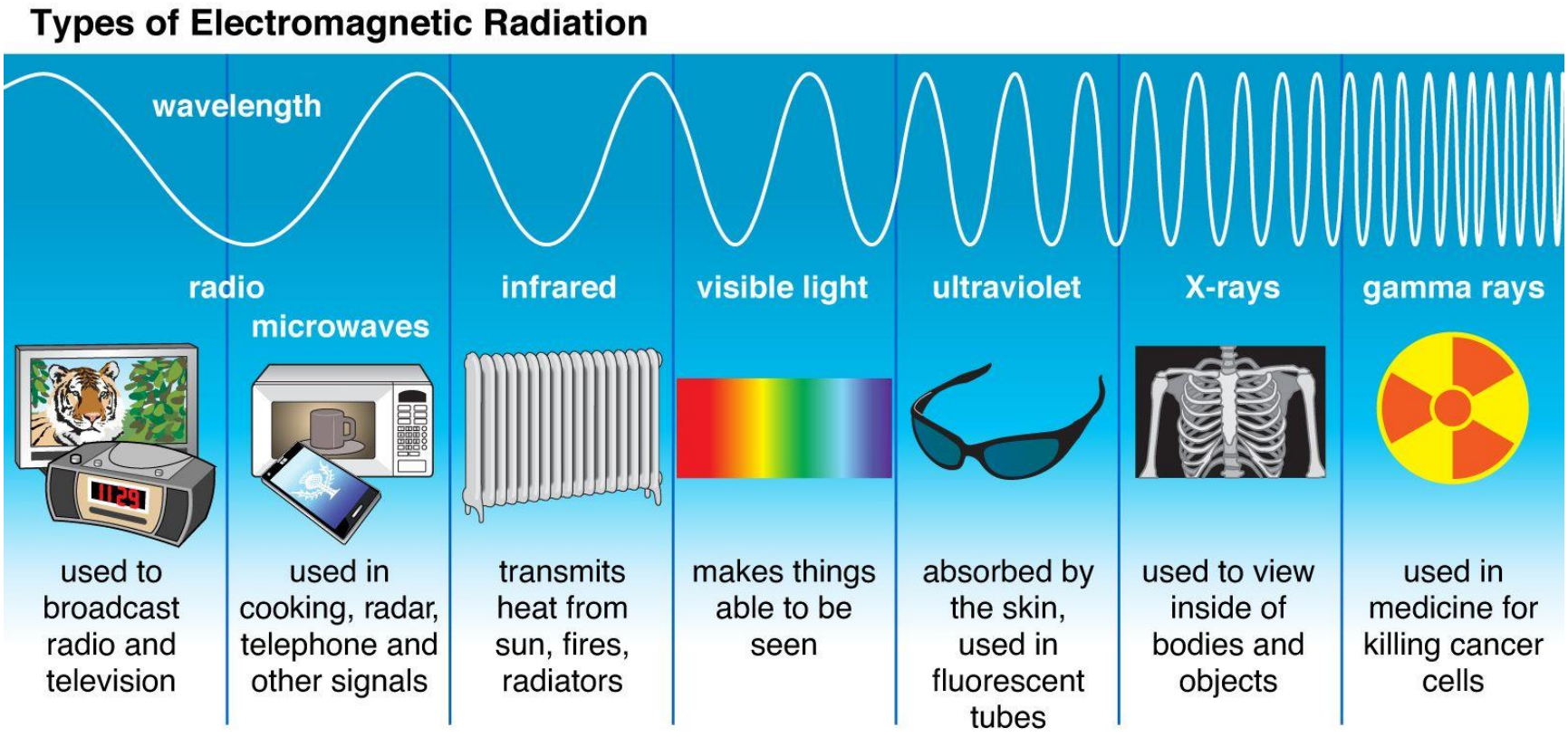




When Ray Optics Fails



EM Waves


Types of Electromagnetic Radiation
wavelength
radio
microwaves
infrared
visible light
ultraviolet
X-rays
gamma rays
used to broadcast radio and television
used in cooking, radar, telephone and other signals
transmits heat from sun, fires, radiators
makes things able to be seen
absorbed by the skin, used in fluorescent tubes
used to view inside of bodies and objects
used in medicine for killing cancer cells
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Maxwell Equation Brief Overview

Maxwell's Equations

Faraday: $$\oint_C \vec{E} \cdot \vec{d\ell} = -\frac{d}{dt} \int_S \vec{B} \cdot \vec{ds}$$

Ampere: $$\oint_C \vec{H} \cdot \vec{d\ell} = \int_S \vec{J} \cdot \vec{ds} + \frac{d}{dt} \int_S \vec{D} \cdot \vec{ds}$$

Gauss - E: $$\oint_S \vec{D} \cdot \vec{ds} = \int_V \rho dv; \quad \vec{D} = \epsilon_0 \vec{E}$$

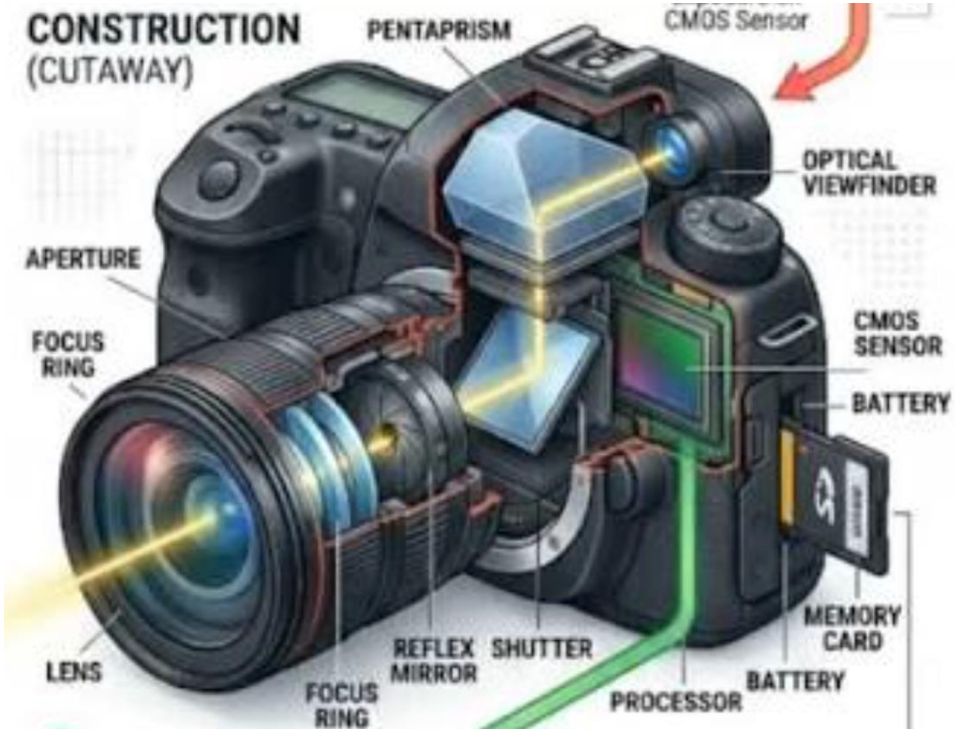
Gauss - B: $$\oint_S \vec{B} \cdot \vec{ds} = 0; \quad \vec{B} = \mu_0 \vec{H}$$

@eeanimation

$$\nabla^2 E - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = 0$$

Problem?

- Why do we need simulations?





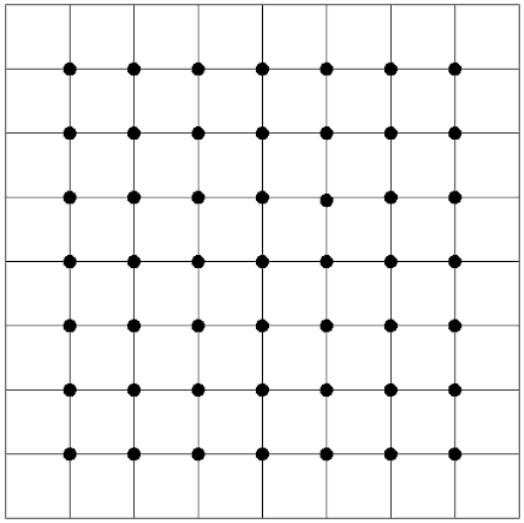
The Reality vs Computer

What computer algorithm did differently from humans?

Computers Cannot Understand Continuous Reality

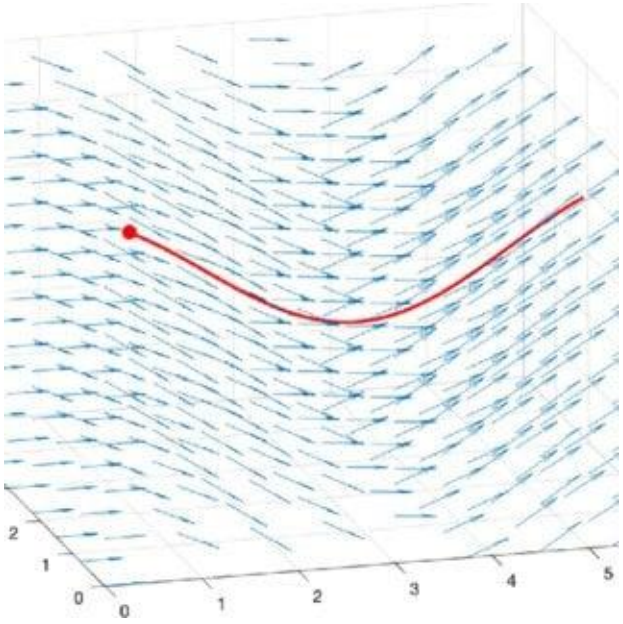
What you see vs What computer sees

$$\mathbf{E}(x, y, z, t)$$



Discretizing Space

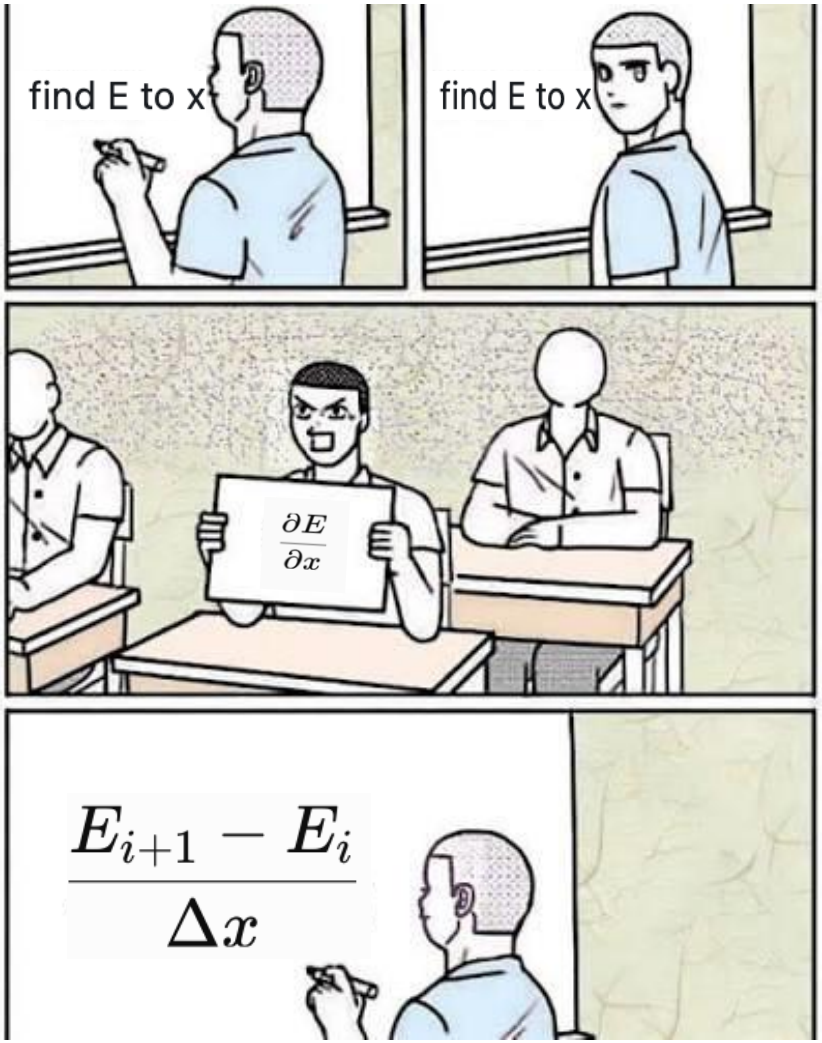
- Make the point into a set of number n points.





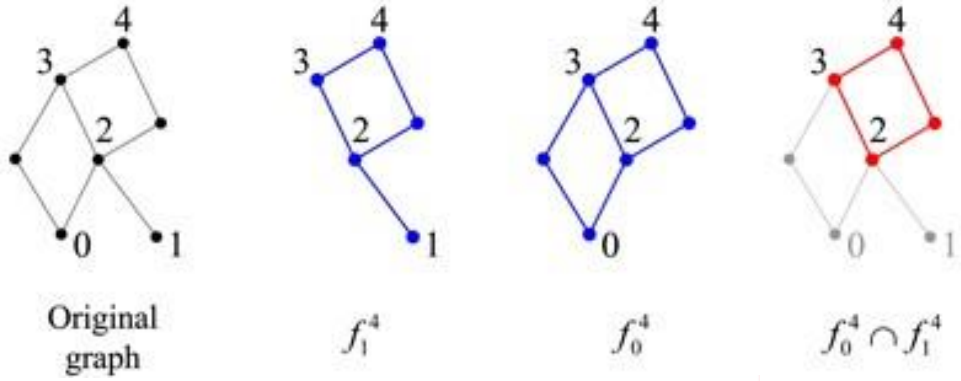
What your computer derives

Your Computer use approximation



Main Differention Method in Computers

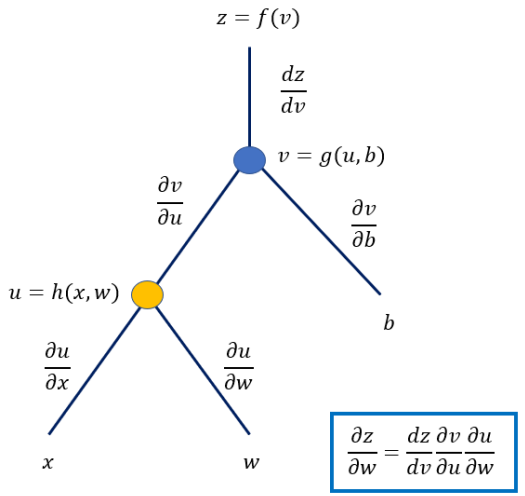
- Symbolic Differentiation



- Numerical Differentiation

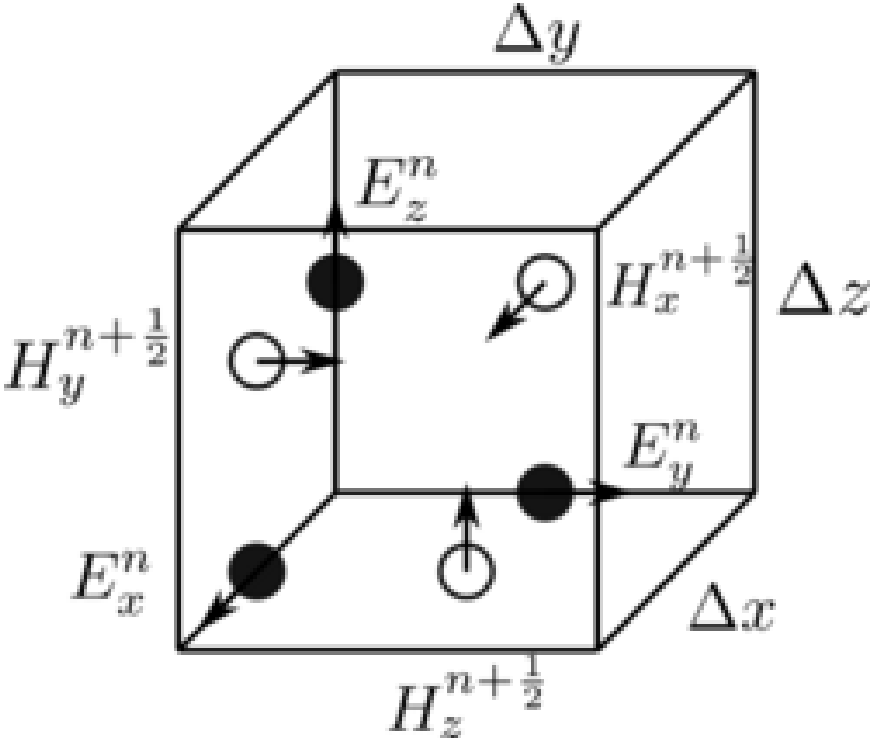
$$f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

- Automatic Differentiation



Finite Difference Time Domain (Yee's Method)

- Electromagnetic field



Update FDTD Algorithm

- Initialize electric field E
- Initialize magnetic field H
- Update H
- Update E
- Apply boundary conditions
- Record results

Boundary Conditions for Simulations

- Perfect Electric Conductor: $E_{tangent} = 0$
- Absorbing Boundary as preset to prevent reflection

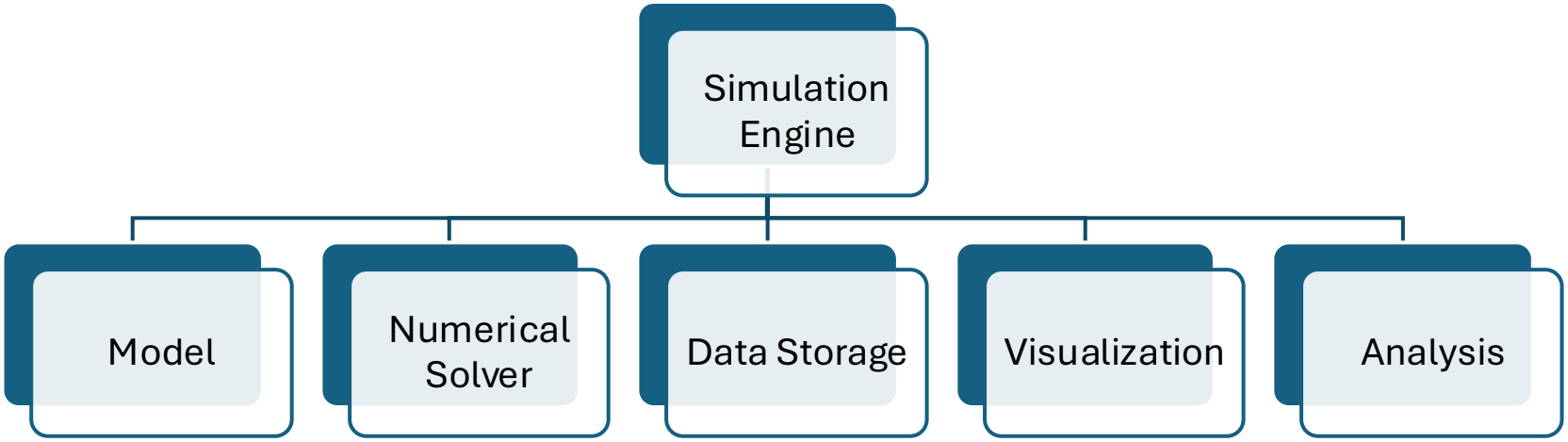
A Correct Formula don’t meant it’s Perfect

- Courant–Friedrichs–Lewy Conditions

$$c\Delta t < \Delta x$$

$$c\Delta t \leq \frac{1}{\sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}$$

Coding Flowchart

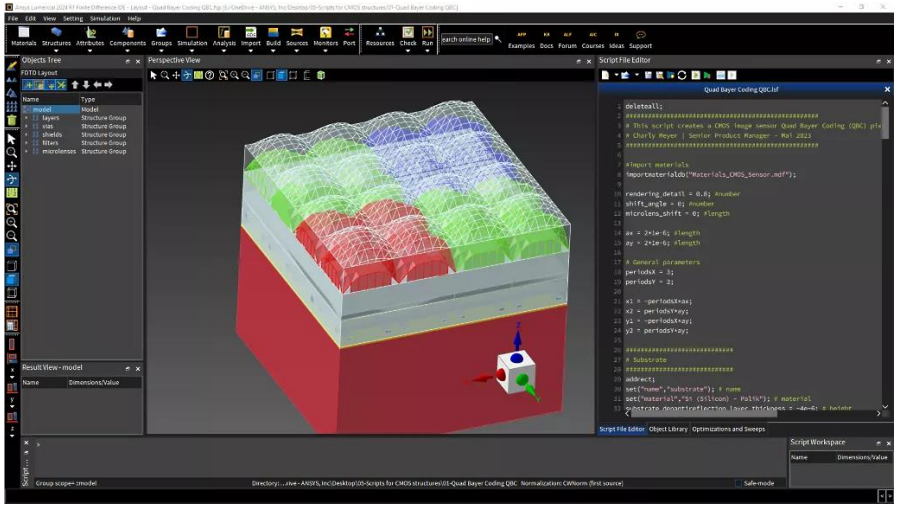


Computational Optics Software

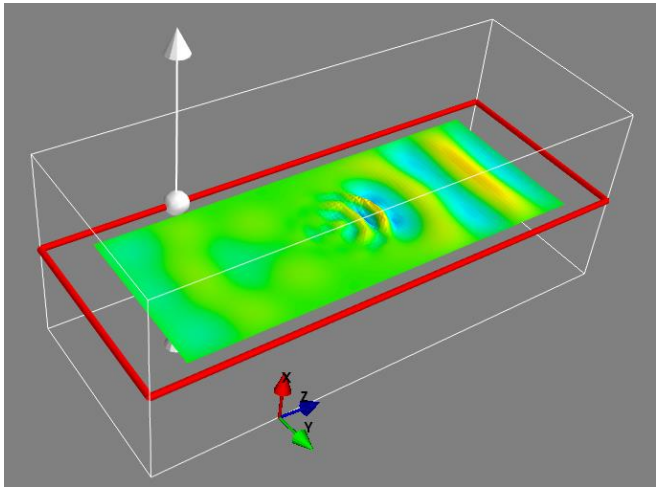
Electromagnetic Simulation	• Numerical Solver
Optical Design	• Tracing / Optimization
Scientific Computation	• Visualization / Analysis

Simulations for Electromagnetics

- ANSYS Lumerical

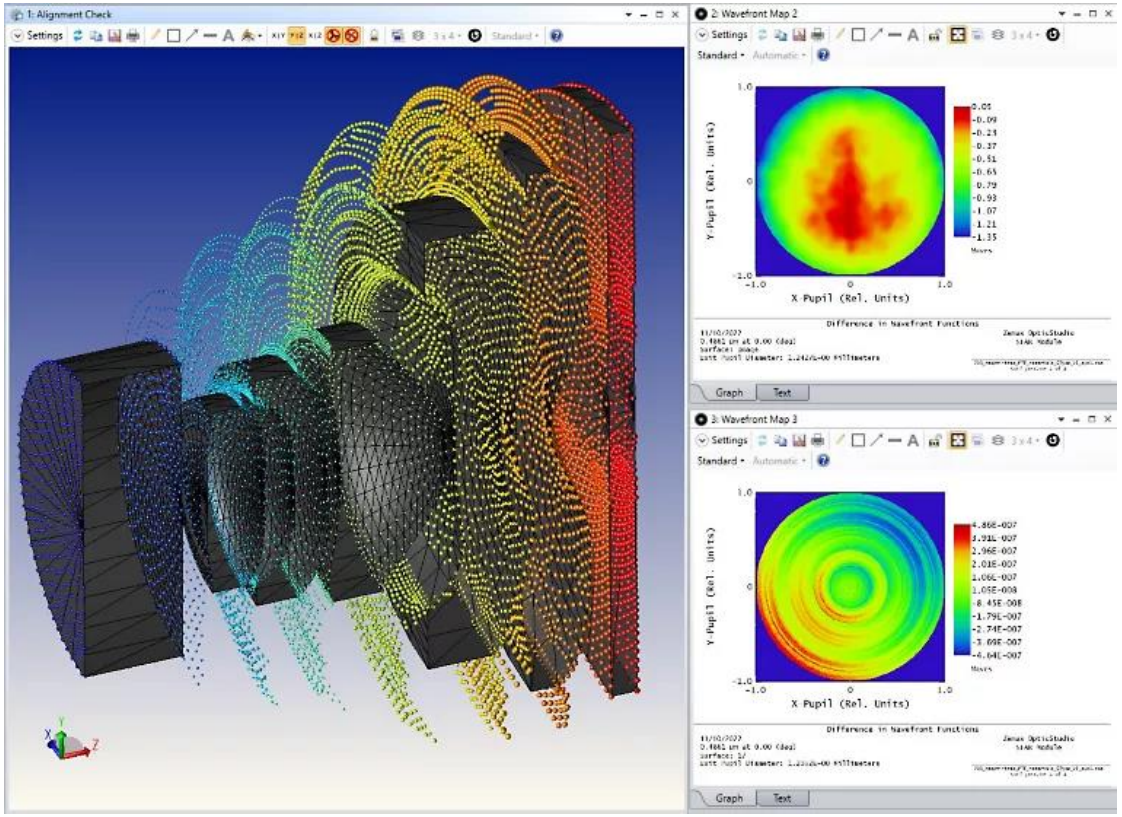


- MEEP



Optical Design Software

- ANSYS Zemax OpticStudio

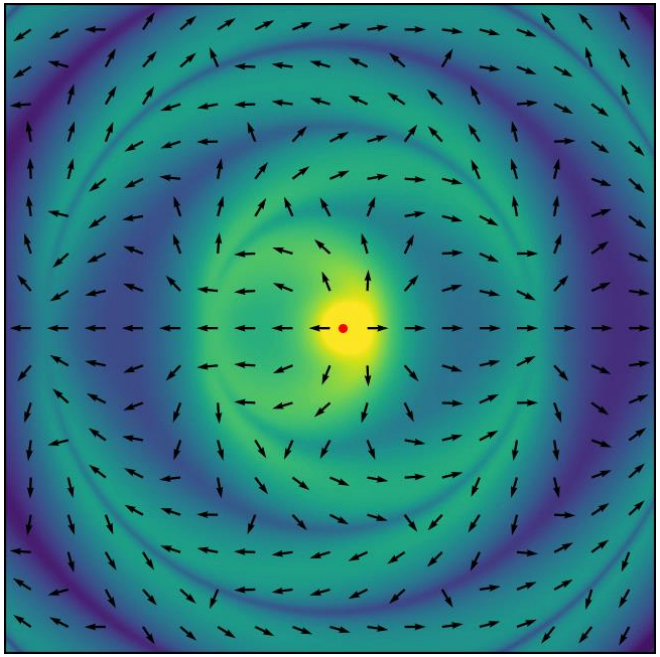


Coding Languages

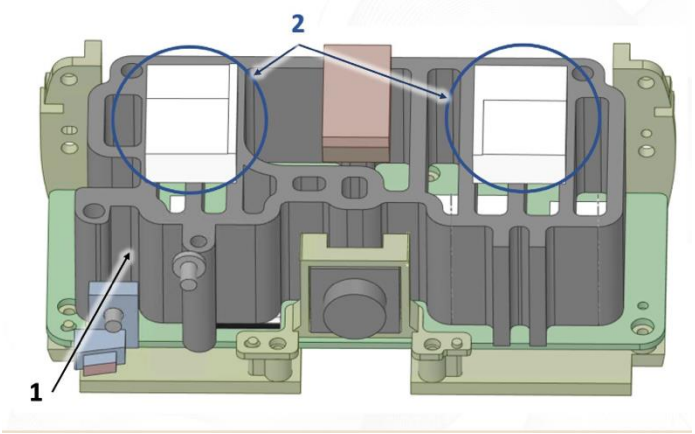


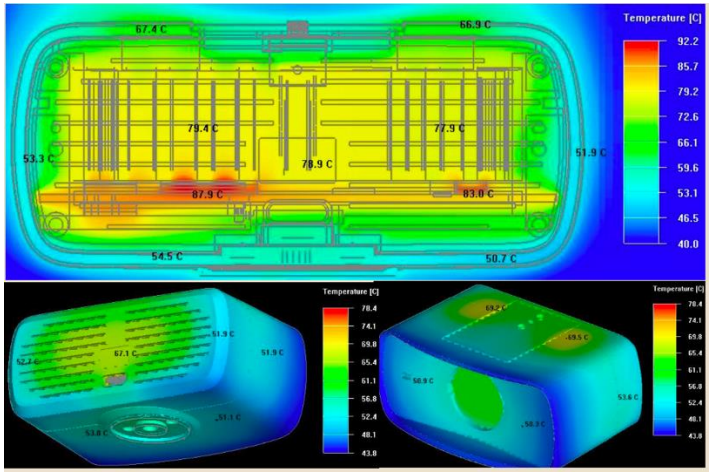
My Experience

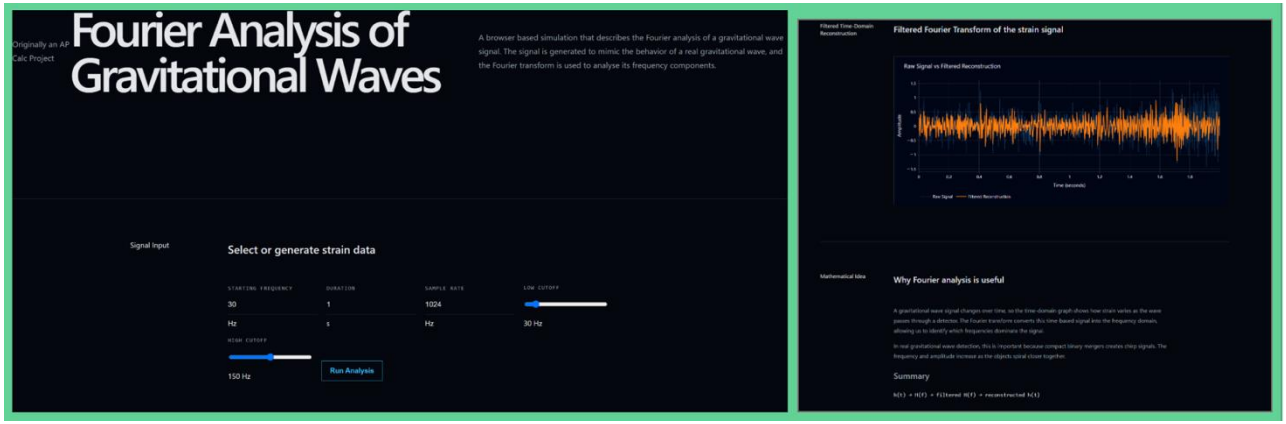




My Experience







Thank you